



Phenylvinilbisquinolines as fluorescent markers in functionalized polypropylene films

Arturo R. Vazquez-Velazquez¹ · Rosa Á. Vazquez-Garcia² · German Hernandez-Bucio³ · Virgilio A. Gonzalez-Gonzalez⁴ · Ivana Moggio⁵ · Sofia Vazquez-Rodriguez⁴ 

Received: 17 January 2019 / Revised: 21 May 2019 / Accepted: 27 May 2019
© Springer-Verlag GmbH Germany, part of Springer Nature 2019

Abstract

In this work, polypropylene films (PP) and maleic anhydride graft polypropylene (PPgMA) at different concentrations of a photoluminescent dye 2,2'-(1,4-phenylenedivinylene) bisquinoline (M1) were prepared by solvent casting. The fluorescence spectrum of M1-dye showed a maximum peak at 424 nm in xylene which corresponds to the blue region of the electromagnetic spectrum. In the PP films, new peaks at 505 and 535 nm appeared that can be ascribed to interchain species formed by intermolecular π - π stacking and likely hydrogen-bonding interaction as found for other quinaldine oligomers. By thermal annealing of the film heating at 130 °C during 60 min, it was possible to overcome these interactions. The results show that thermal annealing of films is able to disrupt M1-aggregates as a consequence of macromolecular reorganization promoted by heating. The emission responses detected in PP/PPgMA/M1 films are associated with a small-dispersed M1-aggregates associated with a polar character between maleic anhydride and M1. The fluorescence changes of M1 in the polymer films indicate the possibility of using this bisquinoline-dye as an optical sensor for detecting structural modification of thermoplastic polymers.

Keywords Fluorescence sensor · Mechanochromism · Polypropylene · Maleic anhydride

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00289-019-02828-6>) contains supplementary material, which is available to authorized users.

✉ Sofia Vazquez-Rodriguez
sofia.vazquezrd@uanl.edu.mx

Extended author information available on the last page of the article

Introduction

Detection of mechanical stress through fluorescent mechanoresponse of a polymer/fluorescent-dye blend is a novel application for detecting premature failure in polymer applications. Fluorescent-dyes are incorporated to a polymer giving a variation of optoelectronic features in response to a stimulus (thermal or mechanical). Different processes can occur during mechanical stimulus such as conformational changes of polymer molecules, dispersion of dye-aggregates in polymer host and scission of covalent and non-covalent bonds [1, 2]. The development of polymer-based mechanochromic materials using aggregation-based response of fluorescent-dye compounds is based on changes of optical properties associated with intermolecular distance and geometric orientation of dye-aggregates [3]. The aggregates are formed because of π - π interactions when the dye is mixed in the polymer host due to phase separation; after applying a mechanical stress, when these are disaggregated, their emission are quenched and the dye recovers its intrinsic emission typical of isolated non-interacting fluorescent molecules [4, 5].

The first example of this sensing mechanism was demonstrated by the group of Weder [6] they used cyano-oligo(p-phenylene vinylene) (cyano-OPV) derivatives in a variety of polymers showing a mechanoresponse that depends on the cyano-OPV concentration and also on the crystallinity of polymer [5, 7]. In the case of polypropylene (PP) as a host matrix of fluorescent-dye, Pucci et al. [8] used bis(benzoxazolyl)stilbene (BBS) incorporated in PP via melt processing for use in deformation sensors. They reported that at higher concentration of BBS (>0.2% BBS), a green emission associated with dye-aggregates could be quenched as a consequence of the aggregates dissociation because of PP tensile orientation [8]. These findings have been used for exploring aggregates properties of terthiophene derivatives [9, 10] and perylene in polyethylene [3, 11] also BBS in polyurethane thermoplastic [12, 13] and biopolymer blends [14].

Recent examples of mechanochromic response have been focused in: detection of damage by surface modification of cellulose nanocrystals using dynamic covalent mechanophore based on a diarylbibenzofuranone [15], self-calibration of mechanochromic response by integration of dispersed chromophores in microcapsules with a mechanic-controlled release of fluorescent-dye-aggregates after mechanical impact, incision or tensile deformation [16] and synthesis of supramolecular chromophores with a mechanoresponsive luminescence and good mechanical performance [17].

In general, the polymer used for this mechanochromic application has to cover certain parameters such as: to be soluble in dye solvents or melt-processable to incorporate the fluorescent-dye, thermoplastic and semicrystalline because the crystalline phase promotes dye-aggregates but the amorphous region provides translational mobility of dyes [2]. In mechanochromic systems, covalent or non-covalent interaction between the fluorescent-dye and the polymer matrix may be observed. In the first case, it is necessary that both components have chemical-related groups, which limits their applications, while the non-covalent interaction

could facilitate a migration of fluorescent-dye to the surface of polymer [1, 18]. In case of PP, the functionalization with maleic anhydride (a polar group) promotes affinity to other polymers changing its adhesion properties. Polypropylene is an important material among polyolefins due to its low density, chemical inertness, low cost, versatility in structural designs with important changes in mechanical properties and viability to design morphological structure by using fillers or blending with other polymers [19]. PP is nowadays a recyclable polymer with high commercial demand suitable in many industrial applications such as automotive, textile, medical equipment, packaging, construction among others [19, 20]. Despite its great advantages, thermochromism and mechanochromism in PP are still limited, which could improve its potential in detecting damages for electric industry appliances (capacitors films and power cables) or in superhydrophobic membranes sensing. In this regard, the effect of grafted maleic anhydride on the mechanochromic response of fluorescent compounds remains unexplored, at least during our research. Recently, findings about this topic was reported by Zich et al., who explored strain response of PE-grafted pyrene through functionalization of PE-grafted maleic anhydride with 1-aminepyrene (AP) [21]. Samples grafted with 1 wt% of AP showed excimer emission onto two PEs with different tensile properties. After mechanical deformation, AP-excimers were disrupted changing emission of polymer film. They found that AP grafted groups were oriented in amorphous phase of PE, therefore the more amorphous the polymer the more dispersion of AP was achieved, which manifested itself in a weak excimer emission.

In this work, the mechanoresponse of dye-aggregates of 2,2'-(1,4 phenylenedivinylene) (M1) bisquinoline in PP matrix is evaluated using 1, 5 and 10 wt% of 1 wt% of maleic anhydride grafted polypropylene (PPgMA) as a aggregate promoter onto PP films. The significant effect of PPgMA on a weak interaction to dye-aggregates of M1 is evaluated by changes in fluorescence-emission of those aggregates after thermal annealing. The dye-aggregates disruption is evaluated at 130 °C while that morphology and thermal properties of PP/PPgMA/M1 films are evaluated in comparison with PP/M1 films at different M1-concentrations. This research constitutes an alternative route to design fluorescence sensor in PP host.

Experimental part

Materials

Polypropylene (Valtec HS013, melt flow index 11 g/10 min, density=0.9 g/cm³, melting temperature 160 °C supplied from Indelpro, Mexico) and polypropylene grafted with 1 wt% of maleic anhydride (Polybond 3200, melt flow index 115 g/10 min, density=0.91 g/cm³, melting temperature 157 °C supplied from Chemtura Corporation) were used as host polymeric matrix. Xylene (≥98.5%) and chloroform (stabilized with amylene, ≥99.9%) were purchased from Fisher Company. M1-dye was synthesized according to the literature procedure [22, 23].

Films preparation and thermal annealing

PP/M1 and PP/PPgMA films with 0.01–1.5 wt% of M1 were prepared by solution casting using xylene as solvent. The concentration of PPgMA in films was 1, 5 and 10 wt%. The polymer pellets and M1-dye were dissolved in xylene, stirred and setting in reflux for 1 h. The solution was casted into Petri dish, and the film was recovered after solvent evaporation. Thermal annealing of film samples was performed, following modified procedures [6, 8] in a thermostatically controlled hot stage at 130 °C during 60 min and immediately quenched by immersion in an ice-water bath at 5 °C. Samples were identified as PP/PPgMA-1, PP/PPgMA-5 and PP/PPgMA-10 when 1, 5 and 10 wt% of maleic anhydride grafted polypropylene was added to film composition.

Characterization

UV–vis spectra were obtained using a PerkinElmer Lambda 35 spectrophotometer and Agilent UV–Vis-NIR Cary 5000, and the absorption spectrum of M1 was determined in xylene solution (4.9×10^{-3} mg/mL) while M1-powder was obtained using an integration sphere at room temperature. The fluorescence spectra of M1 in solution (4.9×10^{-3} mg/mL) and films of PP/M1 and PP/PPgMA/M1 were acquired using a PerkinElmer luminescence spectrometer LS55, by λ_{exc} at 379 nm. For reducing the surface scattering during fluorescence measurement, a drop of poly(methylphenylsiloxane) was used between the polymeric film with the quartz slides to keep them planar and decrease the films roughness [8]. Thermogravimetric analysis of M1 was performed using a Linseis STA PT1600 simultaneous thermogravimetry at 10 °C/min rate from room temperature to 600 °C under air atmosphere. The scanning electron microscopy (SEM) analysis on polymers films was performed with a FEI Nova NanoSEM 200 microscope, operated at an accelerating voltage of 5 kV. Microscopy optic images were obtained using Olympus BX51 M through a 5x objective lens. Films of PP/M1 and PP/PPgMA/M1 at 1.5 wt% of M1 were evaluated by infrared spectroscopy (FTIR) using a spectrophotometer Shimadzu IR Affinity-1s, in transmittance mode at 4 cm^{-1} of resolution with 32 scans. The differential scanning calorimetry (DSC) was performed by using PerkinElmer Diamond DSC equipment with a 10 °C/min rate under nitrogen atmosphere.

Results and discussion

Optical properties of fluorescent compound M1 in solution

The UV–vis absorption spectrum of M1 in xylene solution (Fig. 1a) presents one main absorption peak at 379 nm due to the π – π^* electronic transitions of the conjugated chain [22–24]. The emission spectrum measurement in xylene (Fig. 1a) shows a band at 424 nm associated with the emission of the isolated molecule (monomer)

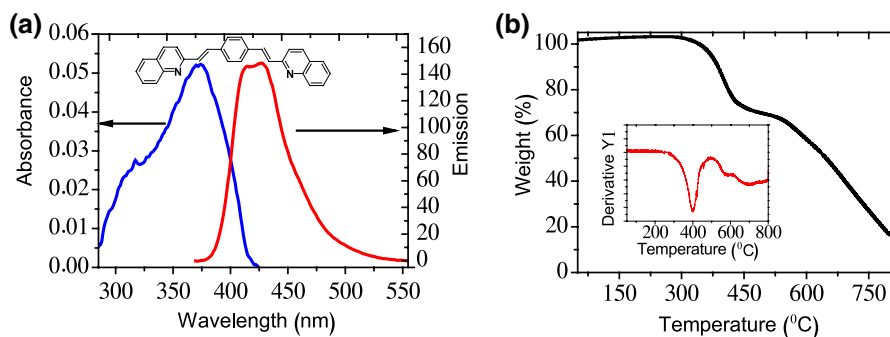


Fig. 1 **a** UV–vis absorption and emission spectra ($\lambda_{\text{exc}}=379$ nm) of 2,2'-(1,4-phenylenedivinylene) bis-quinoline in xylene solution (4.9×10^{-3} mg/mL). **b** Thermogravimetric analysis of M1 under air atmosphere

of M1, as it was reported in chloroform solution [23]. According to the TGA result of M1 (Fig. 1b), the initial degradation temperature ($^{\circ}\text{C}$) and char (%) were found as 275 and 17, which suggests that M1 is thermally stable at temperature of polymer solution mixing (~ 130 $^{\circ}\text{C}$) used in this research.

Characterization of PP/M1 films at different concentrations of M1 and PPgMA

The digital images of PP/M1 and PP/PPgMA-1 films, under the excitation of a long-range UV lamp ($\lambda_{\text{exc}}=365$ nm), containing different concentrations of M1 (0.01–1.5 wt%) are shown in Fig. 2. The PP/M1 films (Fig. 2 PP-M1) show a homogeneous blue color at 0.01, 0.05 and 0.1 wt% of M1 while the films with 0.5, 1.0 and 1.5 wt% present a blue color with yellow–green regions that look like fibrils for 0.5% and spherulites for higher concentrations. Same results were observed for PP/PPgMA-1 films (Fig. 2 PP/PPgMA-1), except for the film at 0.5 wt% of M1 in which a lower level of micro-aggregates of M1 on polymer surface compared to PP/M1 film was observed. The emission spectra for PP/M1 and PP/PPgMA-1/M1 films at 0.01, 0.05 and 0.1 wt% of M1 (Fig. 2) showed a maximum at 424 nm, similarly to that obtained for M1 dissolved in xylene, indicating that the dye is well dispersed in this concentration range. For higher contents, new peaks at 505 and 535 nm appeared that can be ascribed to interchain species formed by intermolecular π – π stacking and likely hydrogen-bonding interaction, as found for other quinaldine oligomers [22, 24].

The morphology of the aggregates is observed clearly by optical microscopy and SEM analysis at 1 wt% of PPgMA and 1.5 wt% of M1 (PP/PPgMA-1/M1-1.5). M1-crystals were growing on the surface of polymer as dendrites and mosaic structures [25] observed by SEM (Fig. 3). The spherulites size of polymer has an average diameter of 34 μm for PP/M1-1.5 (Fig. 3a) and 23 μm for PP/PPgMA-1/M1-1.5 films (Fig. 3c). Furthermore, the crystal size of M1 is larger in PP/PPgMA-1/M1-1.5 (Fig. 3d) than PP/M1 film (Fig. 3b). This is caused for a slow evaporation of solvent promoting a higher formation of spherulites of the polymer allowing M1-aggregate on the surface. SEM images of samples at 5 and 10% of PPgMA (Fig S3 in

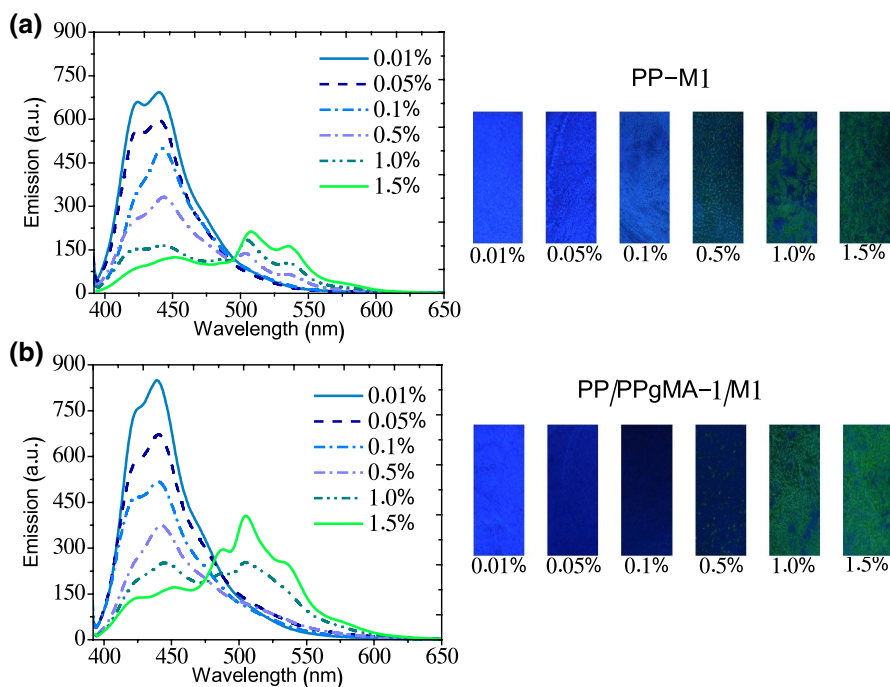


Fig. 2 Digital image and emission spectra of **a** PP/M1 and **b** PP/PPgMA-1/M1 films containing different concentrations of M1 under UV lamp exposure at 365 nm before thermal alignment. ($\lambda_{\text{exc}} = 379$ nm)

supplementary information) are promoting dispersed and larger M1-aggregates onto polymer surface. Thus, the migration of monomer emission of M1 could be restricted, as a monomer as aggregates, by supramolecular interactions between maleic groups of PPgMA and quinoline segment of M1-dye.

The intermolecular interaction was demonstrated by FTIR analysis (in range of $2000\text{--}1500\text{ cm}^{-1}$) of films containing 1.5 wt% of M1 (Fig. 4). In the carbonyl region in all PP/PPgMA spectra, anhydride carbonyl peak absorption at 1785 cm^{-1} (C=O symmetric stretching) almost disappear indicating interaction between maleic groups to quinoline segment of M1. It has been reported that maleic anhydride groups grafted in PP are very reactive to different chemical groups leading covalent bonds [26], but also intermolecular interactions, such as hydrogen bonds, may induce the high affinity for inorganic nanoparticles [27]. In both reported cases, characteristic peaks absorption of maleic anhydride were diminished in FTIR spectra.

Thermal properties of 1.5 wt% M1-dye and different levels of PPgMA in PP were evaluated by DSC (Table 1). According to the results, during the first heating, the crystallinity observed for PP and PP/M1-1.5 increased from 34 to 38%, instead of adding 1, 5 and 10% PPgMA in PP/M1-1.5, the crystallinity of PP film was around 42% for all films. After erasing thermal history, crystallinity for all samples was around 35%, except in PP/PPgMA-1 (33%) and PP/PPgMA-10/M1-1.5 films (37%). Thus, the incorporation of M1 and PPgMA (at 1 and 5 wt%) is not increasing crystallinity of polymer as it has been observed in other ternary mechanochromic blends

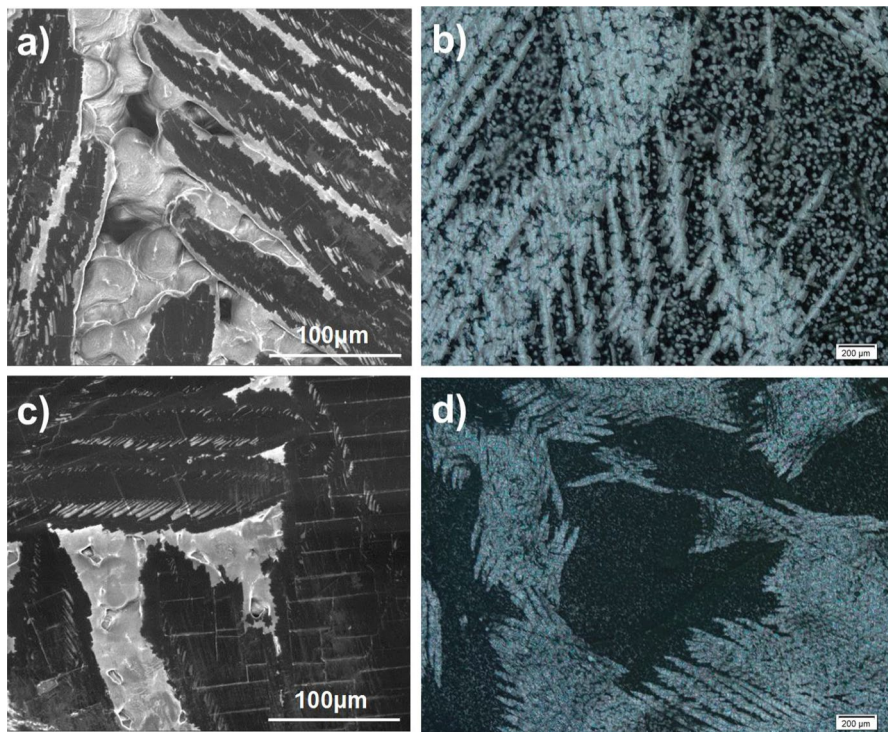


Fig. 3 SEM micrographs and optical microscopy (OM) images of PP/M1 (**a** SEM, **b** OM) and PPgMA-1/M1 (**c** SEM, **d** OM) at 1.5 wt% of M1

Fig. 4 FTIR spectra of PP/PPgMA/M1-1.5 films with different concentrations of PPgMA

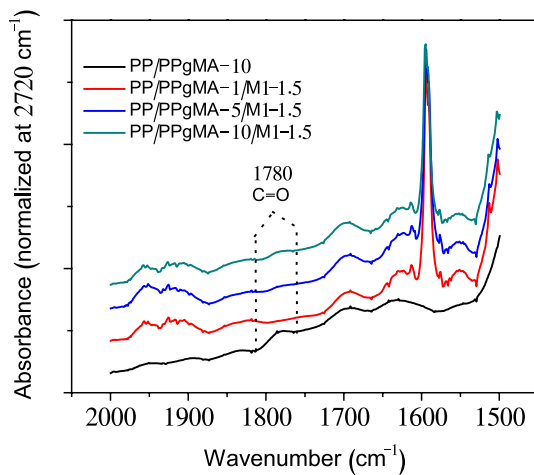


Table 1 Thermal properties of PP and PP/PPgMA films with 1.5 wt% M1

	$T_{c\text{onset}}$ (°C)	Thermal behavior (first heating scan)			Thermal behavior (second heating scan)		
		T_m (°C)	ΔH_m (J/g)	% χ	T_m (°C)	ΔH_m (J/g)	% χ
PP	118	162	69.8	34	158	70.4	34
PP/M1-1.5	121	163	78.0	38	159	71.8	35
PP/PPgMA-1	119	164	86.7	42	155	69.3	33
PP/PPgMA-1/M1-1.5	128	164	84.3	41	161	75.4	36
PP/PPgMA-5/M1-1.5	128	164	87.0	42	161	73.0	35
PP/PPgMA-10/M1-1.5	128	165	90.0	43	162	76.4	37

[14]. It has been reported that the crystallization of fluorescent-dye and its polymer compatibility are relevant for preventing phase segregation at high level of dye concentration which could be manipulated through fast quenching and plasticization of mechanochromic blend [9, 14]. In our research, the incorporation of PPgMA in PP/M1 system is promoting M1-aggregates by intermolecular interaction between functional groups, expecting more concentration of M1-aggregates and increasing emission onto PP/PPgMA film, decreasing a possible migration of M1 to the surface. M1-dye migrated to PP surface after more than 100 days, showing an increase in emission fluorescence (Fig S5 supplementary information), while PP/PPgMA-1/M1-1.5 showed a lower emission.

Effect of thermal annealing of films on optical properties

To determinate the annealing time necessary for breaking the M1-aggregates at 130 °C, films of PP/M1 and PP/PPgMA-1/M1 with 0.5 wt% of M1 were heated at different times (15, 30, 45 and 60 min) monitoring changes in intensity of the dye-aggregate peak emission (505 and 535 nm). The initial emission spectrum

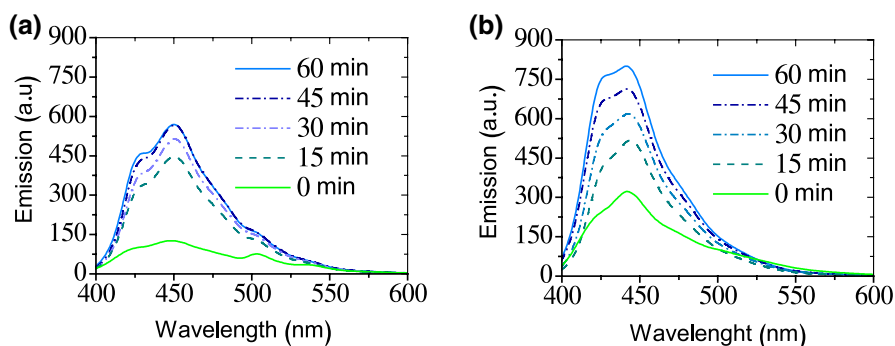


Fig. 5 Emission spectra of **a** PP-M1 and **b** PP/PPgMA-1/M1 with 0.5 wt% of M1 film at different thermal alignment times at 130 °C. ($\lambda_{\text{exc}} = 379$ nm)

(Fig. 5a $t=0$ min) for PP/M1-0.5 film exhibits the aggregates peaks at 505 and 535 nm and the monomeric emission at 424 and 445 nm. After thermal annealing during 15 min, the disruption of M1-aggregates is evident due to an increase in the intensity of the peaks at 424 and 445 nm. Thermal annealing time at 30, 45 and 60 min promoted a slow increase in intensity of monomer M1 peak emission in PP/M1 film compared to the dramatic arise after just 15 min of heating. Instead, the PP/PPgMA-1/M1-0.5 film (Fig. 5b) showed a slow and controllable M1-dye monomer emission at different annealing time besides a higher emission reached at 60 min compared to PP/M1-0.5 film. For this reason, the thermal annealing time used in this work was 60 min.

In the Fig. 6, is shown the emission response of films containing 1.5 wt% of M1 after thermal annealing during 60 min at 130 °C. For a better analysis, the emission spectra of films were normalized at 424 nm (monomer emission of M1). For PP/M1-1.5 (Fig. 6a) and PP/PPgMA-1/M1-1.5 (Fig. 6b) after thermal annealing, it is evident the considerable reduction in M1-dye-aggregates emission at 505 nm and 535 nm. These results show that the thermal-annealing protocol is able to disrupt M1-aggregates as a consequence of macromolecular reorganization promoted by heating. Besides, M1-aggregates emission at 505 nm in PP/PPgMA-1/M1-1.5 film is slightly more intense and shows three shoulders compared to PP/M1-1.5 film. It is important to highlight that concentration higher than 1 wt% of PPgMA increased the intensity of M1-dye-aggregates emission; however, at this thermal annealing conditions, those aggregates were not disrupted easily. As a consequence, segregation to M1-dye monomer emission was not evident (Fig S4 supplementary information) at 5 and 10% of PPgMA.

Even though an optimization of fluorescence onto films was evident using PPgMA, a better dispersed M1-dye could be obtained by melt processing of polymers, such as it was demonstrated in polyethylene with tertiophene derivative dyes [8]. Thus, using modified polyolefins is a route to manipulate emission for mechanochromic polymer system.

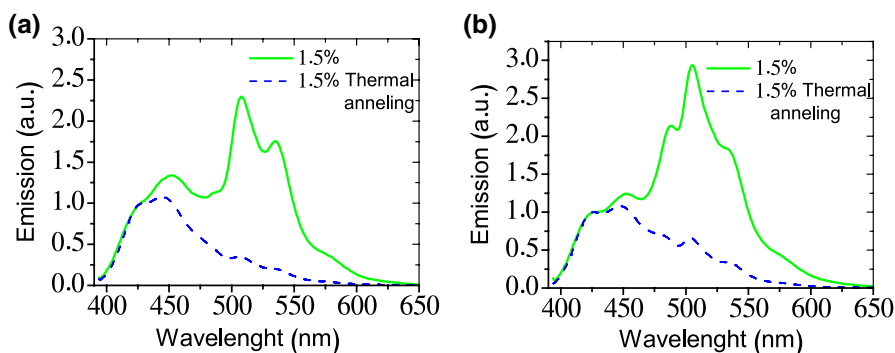


Fig. 6 Emission spectra of **a** PP/M1 and **b** PP/PPgMA-1/M1 at 1.5% of M1 after thermal annealing in vacuum oven at 130 °C during 1 h. ($\lambda_{\text{exc}}=379$ nm). Normalized emission at maximum of the band related to monomer emission at 424 nm

Conclusions

In conclusion, we have described the effect of PPgMA on the interactions with fluorescent M1-dye in PP films. We found that M1-dye has the capability to grow on the polymer surface as crystal and provide of this manner the emission properties of the polymer films. It was observed that the presence of maleic anhydride group brings on a better dispersion of M1-dye in polymer. The change in the emission of the polymer films after thermal annealing indicates that M1-dye shows thermal and mechanochromic features. Thus, M1-dye could be considered as a fluorescent marker in polyolefins.

Acknowledgements A. R. Vázquez-Velázquez thanks to CONACYT-Mexico for Doctoral Fellowship. The authors acknowledge CONACYT for the financial support through CB-240271 project. The authors are thankful to Dra. Sugehidy Carranza-Bernal (UANL) for FTIR analysis, K. Pinedo-Arriaga for her support during microscopy optic observations and L.E. Roman-Quirino for DSC measurements.

References

1. Calvino C, Neumann L, Weder C, Schrettl S (2017) Approaches to polymeric mechanochromic materials. *J Polym Sci, Part A: Polym Chem* 55:640–652. <https://doi.org/10.1002/pola.28445>
2. Pucci A, Ruggeri G (2011) Mechanochromic polymer blends. *J Mater Chem* 21:8282–8291. <https://doi.org/10.1039/c0jm03653f>
3. Pucci A, Donati F, Ruggeri G, Ciardelli F (2009) Linear low density polyethylene (LLDPE) films containing perylene dyes as stress-strain luminescent indicators. *e-Polymers* 9:058. <https://doi.org/10.1515/epoly.2009.9.1.716>
4. Lott J, Weder C (2010) Luminescent mechanochromic sensors based on poly(vinylidene fluoride) and excimer-forming p-phenylene vinylene dyes. *Macromol Chem Phys* 211:28–34. <https://doi.org/10.1002/macp.200900476>
5. Lavrenova A, Holtz A, Simon YC, Weder C (2016) Deformation-induced color changes in melt-processed polyamide 12 blends. *Macromol Mater Eng* 301:549–554. <https://doi.org/10.1002/mame.201500403>
6. Crenshaw BR, Weder C (2003) Deformation-induced color changes in melt-processed photoluminescent polymer blends. *Chem Mater* 15:4717–4724. <https://doi.org/10.1021/cm034447t>
7. Crenshaw BR, Burnworth M, Khariwala D, Hiltner A, Mather PT, Simha R, Weder C (2007) Deformation-induced color changes in mechanochromic polyethylene blends. *Macromolecules* 40(7):2400–2408. <https://doi.org/10.1021/ma062936j>
8. Pucci A, Bertoldo M, Bronco S (2005) Luminescent Bis(benzoxazoly)stilbene as a molecular probe for poly(propylene) film deformation. *Macromol Rapid Commun* 26:1043–1048. <https://doi.org/10.1002/marc.200500227>
9. Pucci A, Tirelli N, Ruggeri G, Ciardelli F (2005) Absorption and emission dichroism of polyethylene films with molecularly dispersed push-pull terthiophenes. *Macromol Chem Phys* 206:102–111. <https://doi.org/10.1002/macp.200400171>
10. Pucci A, Cappelli C, Bronco S, Ruggeri G (2006) Dichroic properties of bis(benzoxazoly)stilbene and bis(benzoxazoly)thiophene dispersed into oriented polyethylene films: a combined experimental and density functional theory approach. *J Phys Chem B* 110:3127–3134. <https://doi.org/10.1021/jp056325p>
11. Cardelli A, Ricci L, Ruggeri G, Borsacchi S, Geppi M (2011) Optical properties of a polyethylene dispersion with a luminescent silica prepared by surface grafting of a perylene derivative. *Eur Polymer J* 47:1589–1600. <https://doi.org/10.1016/j.eurpolymj.2011.05.006>

12. Cellini F, Zhou L, Khapli S, Peterson SD, Porfiri M (2016) Large deformations and fluorescence response of mechanochromic polyurethane sensors. *Mech Mater* 93(145):162. <https://doi.org/10.1016/j.mechmat.2015.10.013>
13. Bao S, Li J, Lee KI, Shao S, Hao J, Fei B, Xin JH (2013) Reversible mechanochromism of a luminescent elastomer. *ACS Appl Mater Interfaces* 5:4625–4631. <https://doi.org/10.1021/am4013648>
14. Pucci A, Signori F, Bizzarri R, Bronco S, Ruggeri G, Ciardelli F (2010) Threshold temperature luminescent indicators from biodegradable poly(lactic acid)/poly(butylene succinate) blends. *J Mater Chem* 20:5843. <https://doi.org/10.1039/C0JM00057D>
15. Imato K, Natterodt JC, Sapkota J, Goseki R, Weder C, Takahara A, Otsuka H (2017) Dynamic covalent diarylbibenzofuranone-modified nanocellulose: mechanochromic behaviour and application in self-healing polymer composites. *Polymer Chemistry* 8:2115–2122. <https://doi.org/10.1039/c7py00074j>
16. Calvino C, Guha A, Weder C, Schrettl S (2018) Self-calibrating mechanochromic fluorescent polymers based on encapsulated excimer-forming dyes. *Adv Mater* 30:1704603. <https://doi.org/10.1002/adma.201704603>
17. Lavrenova A, Balkenende DWR, Sagara Y, Schrettl S, Simon YC, Weder C (2017) Mechano- and thermoresponsive photoluminescent supramolecular polymer. *J Am Chem Soc* 139:4302–4305. <https://doi.org/10.1021/jacs.7b00342>
18. Fleischmann C, Lievenbrück M, Ritter H (2015) Polymers and dyes: developments and applications. *Polymers* 7:717–746. <https://doi.org/10.3390/polym7040717>
19. Maddah HA (2016) Polypropylene as a promising plastic: a review. *Am J Polym Sci* 6:1–11. <https://doi.org/10.5923/j.ajps.20160601.01>
20. AL-Oqla FM, Sapuan SM, Ishak MR, Nuraini AA (2016) A decision-making model for selecting the most appropriate natural fiber: polypropylene-based composites for automotive applications. *J Compos Mater* 50:543–556. <https://doi.org/10.1177/0021998315577233>
21. Zych A, Verdelli A, Soliman M, Pinalli R, Pedrini A, Vachon J, Dalcanale E (2019) Strain-reporting pyrene-grafted polyethylene. *Eur Polym J* 111:69–73. <https://doi.org/10.1016/j.eurpolymj.2018.12.016>
22. Liang F, Chen J, Cheng Y, Wang L, Ma D, Jing X, Wang F (2003) Synthesis, characterization, photoluminescent and electroluminescent properties of new conjugated 2,2'-(arylenedivinylene)bis-8-substituted quinolines. *J Mater Chem* 13:1392–1399. <https://doi.org/10.1039/b210408c>
23. Gutiérrez AR et al (2010) Síntesis y mecanosíntesis de fenilendivinilenbisquinolinas para la optoelectrónica. *Superficies y Vacío* 23:73–79. http://smcsyv.fis.cinvestav.mx/supyv23_S/SV23S7310.pdf
24. Gutiérrez AR et al (2015) Mechanosynthesis of a phenylenedivinylidenebisquinoline. Optical, morphological and electroluminescence properties. *J Mol Struct* 1086:138–145. <https://doi.org/10.1016/j.molstruc.2015.01.019>
25. Gránásy L, Pusztai T, Douglas JF (2013) Insights into polymer crystallization from phase-field theory. In: Palsule S (ed) *Encyclopedia of polymers and composites*. Springer, Berlin, pp 1–35. https://doi.org/10.1007/978-3-642-37179-0_30-1
26. Vazquez-Rodriguez S, Sanchez-Valdes S, Rodriguez-Gonzalez FJ, Gonzalez-Cantu MC (2007) Influence of low-molecular-weight diamines in the direct imidation of poly(propylene)-grafted maleic anhydride by melt reaction. *Macromol Mater Eng* 292:1012. <https://doi.org/10.1002/mame.200700086>
27. Liu W, Cheng L, Liu X, Liu C, Chi X, Li S (2018) Correlation between morphology and electrical breakdown strength of the polypropylene/maleic anhydride grafted polypropylene/nano-ZrO₂ ternary system. *J Appl Polym Sci* 135:46842. <https://doi.org/10.1002/app.46842>

Affiliations

Arturo R. Vazquez-Velazquez¹ · Rosa Á. Vazquez-Garcia² · German Hernandez-Bucio³ · Virgilio A. Gonzalez-Gonzalez⁴ · Ivana Moggio⁵ · Sofia Vazquez-Rodriguez⁴ 

- ¹ Centro de Investigación en Materiales Avanzados S.C, Unidad Monterrey, Alianza Norte No. 202, Parque PIIT, 66628 Apodaca, Nuevo León, Mexico
- ² Área Académica de Ciencias de la Tierra y Materiales, Universidad Autónoma del Estado de Hidalgo, Ciudad del Conocimiento, 42184 Pachuca, Hidalgo, Mexico
- ³ División de Ingeniería en Nanotecnología, Instituto Tecnológico Superior de Ciudad Hidalgo, Av. Ing. Carlos Rojas Gutiérrez 2120, Fracc. Valle De La Herradura, 61100 Ciudad, Hidalgo, Mexico
- ⁴ Facultad de Ingeniería Mecánica y Eléctrica, Universidad Autónoma de Nuevo León, Av. Universidad s/n Ciudad Universitaria, 66455 San Nicolás De Los Garza, Nuevo León, Mexico
- ⁵ Centro de Investigación en Química Aplicada (CIQA), Blvd. Enrique Reyna 140, San José de los Cerritos, 25294 Saltillo, Coahuila, Mexico